

Power BI Guide for Education Platform Project

Project Goal

Build an end-to-end Power BI analytics solution for the Education Platform Enterprise System.

This guide covers:

- Data modeling
 - Power Query ETL
 - Star schema design
 - DAX measures
 - Dashboard creation
 - KPI tracking
 - Drill-through reports
 - Incremental refresh concepts
 - Real-world analytics use cases
-

1. Business Understanding

The system manages:

- Leads & inquiries
- Student conversions
- Demo programs
- Batch management
- Attendance
- Tests
- Mock interviews
- Placements
- Trainers
- Companies

The Power BI solution should answer:

- Which lead source converts best?
 - Which trainer performs best?
 - Which course has highest enrollment?
 - What is attendance trend?
 - What is placement success rate?
 - Which company hires most students?
-

2. Power BI Architecture

```
CSV Files / MySQL
  ↓
Power Query ETL
  ↓
Data Cleaning
  ↓
Star Schema Modeling
  ↓
DAX Measures
  ↓
Dashboards & Reports
  ↓
Publishing to Power BI Service
```

3. Recommended Folder Structure

```
PowerBI/
|
|— Dataset/
|   |— inquiries.csv
|   |— students.csv
|   |— batches.csv
|   |— batch_attendance.csv
|   |— tests.csv
|   |— test_results.csv
|   |— interviews.csv
|   |— offers.csv
|
|— PBIX/
|   |— education_platform.pbix
|
|— Documentation/
|   |— data_dictionary.xlsx
```

4. Data Loading

Option 1: Load CSV Files

Steps

1. Open Power BI Desktop
2. Click:

Home → Get Data → Folder

1. Select dataset folder
 2. Transform Data
-

5. Power Query ETL

Recommended Transformations

inquiries

Tasks

- Change data types
 - Trim spaces
 - Remove duplicates
 - Standardize status values
 - Create inquiry month column
-

students

Tasks

- Remove invalid emails
 - Standardize names
 - Create student age group
-

batch_attendance

Tasks

- Convert attendance status
- Create attendance flag

Example:

```
if [status] = "present" then 1 else 0
```

test_results

Tasks

Create grade column.

Example:

```
if [marks] >= 75 then "A"  
else if [marks] >= 60 then "B"  
else "C"
```

6. Recommended Star Schema

FACT TABLES

fact_inquiries

Contains:

- inquiries

Measures:

- total inquiries
- conversions
- lead source metrics

fact_attendance

Contains:

- batch_attendance

Measures:

- attendance percentage
 - absentee rate
-

fact_tests

Contains:

- test_results

Measures:

- average marks
 - pass rate
-

fact_placements

Contains:

- interviews
- offers

Measures:

- placement rate
 - acceptance rate
-

DIMENSION TABLES

dim_students

Contains:

- students

dim_courses

Contains:

- courses
-

dim_trainers

Contains:

- users
-

dim_batches

Contains:

- batches
-

dim_companies

Contains:

- companies
-

dim_date

Create calendar table.

7. Relationships

Recommended Relationships

```
students.student_id
  ↓
batch_enrollments.student_id
```

```
courses.course_id
    ↓
inquiries.course_id

batches.batch_id
    ↓
batch_attendance.batch_id
```

Use:

- One-to-many relationships
- Single direction filtering
- Star schema only

8. Calendar Table

DAX Calendar Table

```
Calendar =
ADDCOLUMNS(
    CALENDAR(DATE(2024,1,1), DATE(2027,12,31)),
    "Year", YEAR([Date]),
    "Month", FORMAT([Date], "MMMM"),
    "MonthNo", MONTH([Date]),
    "Quarter", "Q" & FORMAT([Date], "Q")
)
```

9. Important DAX Measures

CRM Measures

Total Inquiries

```
Total Inquiries = COUNT(inquiries[inquiry_id])
```

Converted Students

```
Converted Students =  
CALCULATE(  
    COUNT(inquiries[inquiry_id]),  
    inquiries[status] = "converted"  
)
```

Conversion Rate

```
Conversion Rate =  
DIVIDE([Converted Students], [Total Inquiries])
```

Attendance Measures

Total Attendance

```
Total Attendance =  
SUM(batch_attendance[attendance_flag])
```

Attendance Percentage

```
Attendance % =  
DIVIDE(  
    SUM(batch_attendance[attendance_flag]),  
    COUNT(batch_attendance[attendance_id])  
)
```

Assessment Measures

Average Marks

```
Average Marks = AVERAGE(test_results[marks])
```

Pass Rate

```
Pass Rate =  
DIVIDE(  
    CALCULATE(COUNT(test_results[result_id]), test_results[marks] >= 40),  
    COUNT(test_results[result_id])  
)
```

Placement Measures

Total Interviews

```
Total Interviews = COUNT(interviews[interview_id])
```

Offers Released

```
Offers Released = COUNT(offers[offer_id])
```

Offer Acceptance Rate

```
Offer Acceptance Rate =  
DIVIDE(  
    CALCULATE(COUNT(offers[offer_id]), offers[status] = "accepted"),  
    COUNT(offers[offer_id])  
)
```

Trainer Measures

Avg Trainer Attendance

```
Avg Trainer Attendance =  
AVERAGE(trainer_metrics[avg_attendance])
```

10. Recommended Dashboards

Dashboard 1: CRM Dashboard

KPIs

- Total inquiries
- Converted students
- Conversion rate
- Lead source contribution
- Counselor performance

Visuals

- KPI cards
 - Funnel chart
 - Pie chart
 - Line chart
 - Table
-

Dashboard 2: Training Dashboard

KPIs

- Total students
- Active batches
- Attendance percentage
- Average marks

Visuals

- Attendance trend

- Heatmap
 - Batch-wise performance
 - Course-wise enrollments
-

Dashboard 3: Placement Dashboard

KPIs

- Total interviews
- Placement rate
- Offer acceptance rate
- Highest hiring company

Visuals

- Company-wise placements
 - Salary distribution
 - Placement trend
 - Student placement status
-

Dashboard 4: Trainer Dashboard

KPIs

- Avg attendance
- Student feedback
- Conversion metrics

Visuals

- Trainer leaderboard
 - Attendance trends
 - Batch performance
-

11. Drill-through Reports

Recommended Drill-through Pages

Student Detail Page

Shows:

- Attendance
 - Tests
 - Mock interviews
 - Placement history
-

Trainer Detail Page

Shows:

- Batch history
 - Student counts
 - Attendance performance
-

Company Detail Page

Shows:

- Open roles
 - Selected students
 - Salary trends
-

12. Recommended Slicers

Use slicers for:

- Course
- Batch
- Trainer
- Company
- Date
- Lead source
- City
- Placement status

13. Power Query Best Practices

Use Staging Queries

```
Raw CSV
↓
Staging Query
↓
Transformation Query
↓
Final Fact Table
```

Disable Load for Intermediate Queries

Improves performance.

Use Proper Data Types

Very important for:

- DAX speed
 - Compression
 - Model size
-

14. Performance Optimization

Best Practices

- Use star schema
 - Avoid many-to-many
 - Reduce calculated columns
 - Use measures instead
 - Hide unused columns
 - Use integer keys
 - Disable auto date/time
-

15. Incremental Refresh Concept

Useful for:

- attendance data
- inquiries
- placements

Instead of refreshing all data:

Old data remains
Only new data refreshes

16. Row-Level Security (RLS)

Example Use Cases

Trainer

Can see only their batches.

Counselor

Can see only assigned leads.

Admin

Can see everything.

Example RLS Filter

```
[trainer_id] = USERPRINCIPALNAME()
```

17. Publishing to Power BI Service

Steps

1. Save PBIX
 2. Click Publish
 3. Select Workspace
 4. Configure refresh schedule
 5. Configure gateway
-

18. Suggested Advanced Features

AI Features

- Student dropout prediction
 - Lead conversion prediction
 - Placement prediction
 - Attendance anomaly detection
-

19. Suggested Data Warehouse Design

Bronze Layer

Raw CSV data

Silver Layer

Cleaned tables

Gold Layer

Analytics-ready star schema

20. Suggested End-to-End Flow

CSV/MySQL

↓

Power Query ETL
↓
Star Schema
↓
DAX Measures
↓
Dashboards
↓
Power BI Service
↓
Scheduled Refresh
↓
Business Users

21. Recommended Visuals

Requirement	Visual
KPI	Card
Trend	Line Chart
Comparison	Bar Chart
Conversion	Funnel Chart
Contribution	Pie/Donut
Geography	Map
Detailed Data	Table/Matrix
Correlation	Scatter Plot

22. Suggested Portfolio Projects

Beginner

- CRM dashboard
- Attendance dashboard

Intermediate

- Placement analytics
 - Trainer performance dashboard
-

Advanced

- Real-time analytics
 - Incremental refresh
 - Data warehouse
 - AI insights dashboard
-

23. Real-World Interview Questions

You can now answer:

- How do you design star schema?
 - Difference between fact and dimension?
 - Why use measures over calculated columns?
 - What is filter context?
 - How does Power Query work?
 - What is incremental refresh?
 - How do you optimize Power BI models?
 - How do you implement RLS?
-

24. Final Learning Outcomes

After completing this project you will understand:

- Power BI ETL
 - Data modeling
 - Star schema
 - DAX
 - Power Query
 - Dashboard design
 - Analytics engineering
 - Performance tuning
 - Enterprise reporting
 - Data warehouse concepts
-

Final Enterprise Workflow

```
graph TD; A[Lead Created] --> B[Inquiry Managed]; B --> C[Demo Attended]; C --> D[Student Converted]; D --> E[Batch Training]; E --> F[Attendance Tracking]; F --> G[Assessments]; G --> H[Mock Interviews]; H --> I[Placements]; I --> J[Power BI Analytics];
```

Lead Created

↓

Inquiry Managed

↓

Demo Attended

↓

Student Converted

↓

Batch Training

↓

Attendance Tracking

↓

Assessments

↓

Mock Interviews

↓

Placements

↓

Power BI Analytics